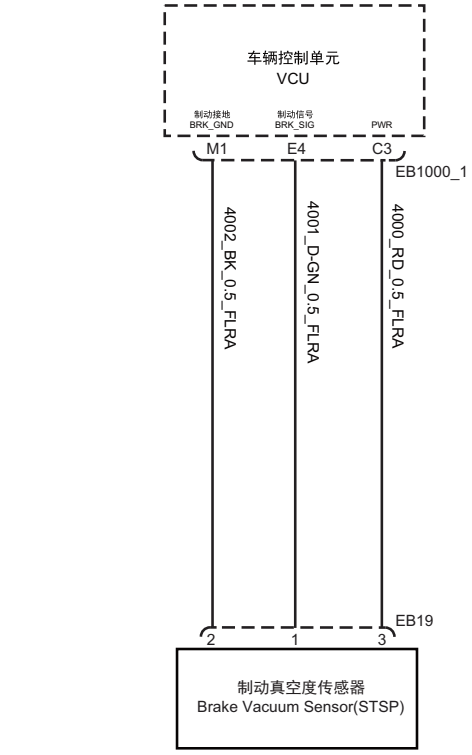
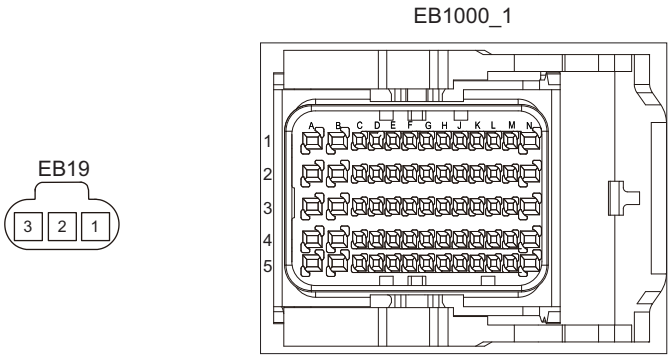
P0558、 P0557、 P0556、 P1C70

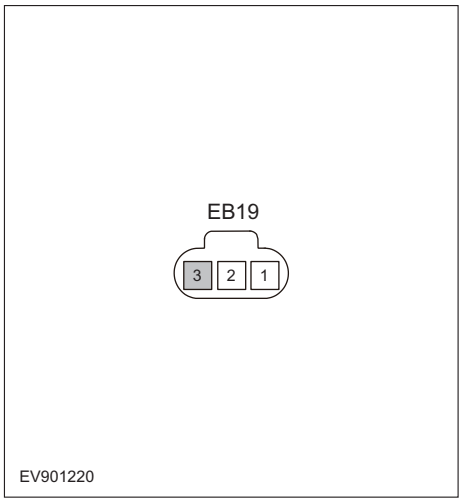
Fault diagnosis code
P0558：DEM_stBrkVacPrsSenSBCfmd
P0557：DEM_stBrkVacPrsSenSGCfmd
P0556：DEM_stBrkVacPrsSenOrng
P1C70：DEM_stBrkVacPrsSenPlaus
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • VCU module fault • The brake vacuum sensor fault.
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P0558
Brake vacuum sensor voltage is greater than or equal to 4.9V
P0557
Brake vacuum sensor voltage is lower than or equal to 4.9V
P0556
Any of the following conditions are satisfied: 1. Brake vacuum sensor voltage is greater than 4.5V ; 2, Brake vacuum sensor voltage is lower than 4.9V ;
P1C70
The diagnosis has been disabled
To set the effect of a fault code condition
Description of circuit diagram
The VCU module monitors the brake vacuum sensor via low level reference circuit and signal circuit.

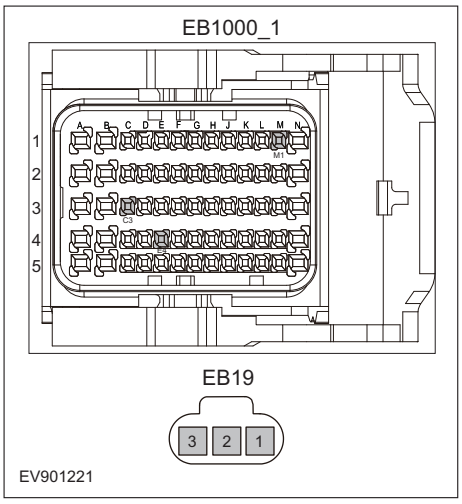
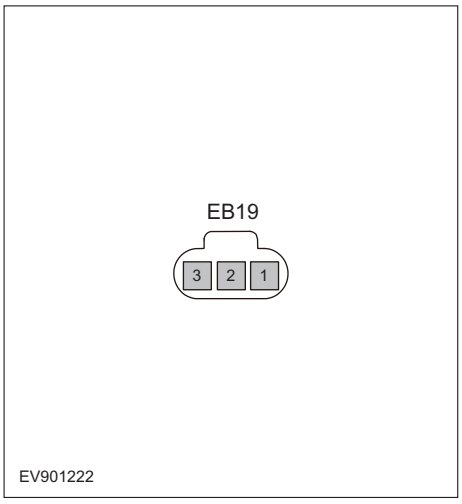
Circuit diagram

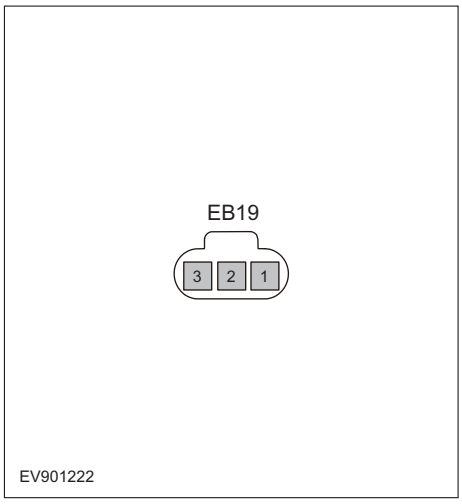




EV901203

Troubleshooting steps	
Step 1	General inspection.
	1 Check the fuse for damage. 2 Check battery capacity. 3 Check the harness plug of brake vacuum sensor,VCU for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 2	Check fault codes.
	1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the VCU with diagnostic apparatus. 4 Is there any other fault code except for P0558、 P0557、 P0556、 P1C70?
	YES → Refer to:DTC Summary list(VCU).
NO ↓	
Step 3	Check the supply power of brake vacuum sensor when it is started or operating.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB19 of brake vacuum sensor. 3 Turn the ignition switch to ON. 4 Measure the voltage between the plug terminal of brake vacuum sensor and ground. Measuring circuit: voltage between terminal 3 on plug EB19 and ground. **Standard value:4.7~5.2V** 5 Check whether the result is normal or not?
	NO → Repair the circuit.
YES ↓	

Step 4	Check the power supply circuit of brake vacuum sensor circuit(open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB1000_1 of GAW module. 3 Measure the resistance between the harness terminal of brake vacuum sensor and the VCU. Measuring circuit: resistance between terminal 1 on plug EB19 and terminal E4 on plug EB1000_1. Measuring circuit: resistance between terminal 2 on plug EB19 and terminal M1 on plug EB1000_1. Measuring circuit: resistance between terminal 3 on plug EB19 and terminal C3 on plug EB1000_1. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 5	Check the power supply circuit of brake vacuum sensor circuit(short circuit).
	1 Turn the ignition switch to OFF. 2 Measure the resistance between the harness plug of brake vacuum sensor and ground. Measuring circuit: resistance between terminal 1 on plug EB19 and ground. Measuring circuit: resistance between terminal 2 on plug EB19 and ground. Measuring circuit: resistance between terminal 3 on plug EB19 and ground. **Standard value: ∞** 3 Measure the voltage between the plug terminal of brake vacuum sensor and ground. Measuring circuit: voltage between terminal 1 on plug EB19 and ground. Measuring circuit: voltage between terminal 2 on plug EB19 and ground. Measuring circuit: voltage between terminal 3 on plug EB19 and ground. **Standard value: ≈ 0V** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Check the brake vacuum sensor (components).
	1 Measure the resistance of the terminal of brake vacuum sensor directly. Measuring circuit: resistance between terminal 1 and terminal 2 on plug EB19. **Standard value:** 2 Measure the resistance between the brake vacuum sensor terminal and the housing. Measuring circuit: resistance between terminal 1 on plug EB19 and the housing. Measuring circuit: resistance between terminal 2 on plug EB19 and the housing. Measuring circuit: resistance between terminal 3 on plug EB19 and the housing. **Standard value: ∞** 3 Check whether the result is normal or not?
	NO → Replace brake vacuum sensor.
YES ↓	
Step 7	Test whether VCU is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB1000_1, IP55_A of VCU. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the VCU plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Change the VCU Module
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.	
Next Step ↓	
Diagnosis end.	